MTH2003 Tutorial Sheet 1

Please attempt all of the questions before your week 3 tutorial, so that you can contribute to and benefit from the discussion.

1. Determine whether the first order ordinary differential equation

$$ty\frac{\mathrm{d}y}{\mathrm{d}t} = t^2 + y^2$$

can be solved:

[i] by the method of separation of variables,
[ii] as a homogeneous first-order ODE,
[iii] as a linear first-order ODE using an integrating factor.

For each method that is applicable, find the solutions and identify the intervals on which your solutions are defined.

2. A body of constant mass is launched vertically upwards from the Earth's surface, with an initial velocity $v_0 > 0$. The body is in passive flight (i.e. there is no propeller or jet engine, etc). A simplified model that neglects air resistance, Earth's rotation, attraction by the Moon, cosmic wind, etc., leads to the following ODE relating the elevation x of the body above the starting point and its velocity v (which is positive if the body is moving upwards):

$$\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{gR^2}{v(R+x)^2}\,,$$

where $R > 0$ is the Earth's radius and $g > 0$ is the acceleration due to gravity.

(a) Find the solution $v(x)$ of this ODE satisfying the given initial condition.
(b) Analyse the interval on which this solution is defined, and the dependence of this interval on v_0. Hence determine:

- the expression for the minimal initial velocity $v_0 = v_*$ that is required in order that the body escapes Earth's gravitation forever. Also evaluate the numerical value to 3 s.f. for $R = 6371\,\mathrm{km}$ and $g = 9.81\,\mathrm{m/s^2}$;
- the velocity of the body at the infinite distance from the Earth, $v_\infty = v(x \to \infty)$, if $v_0 > v_*$;
- the maximal altitude x_* the body reaches if $v_0 < v_*$.

(c) Sketch a representative sample of integral curves according to the following instructions:

- Assume that the units of measurement are chosen so that $R = 1$ and $g = 1$; use the limits $x \in [-1, 2]$, $v \in [-4, 4]$ for the plot;
- Plot the graphs of the solutions corresponding to $v_0 = 0$, $v_0 = v_1$, $v_0 = v_2$ and $v_0 = v_*$, where you need to choose v_1 and v_2 so that $v_1 < v_*$ and $v_2 > v_*$. You may wish to use different colours to draw graphs corresponding to different v_0.
- Plot also the segments corresponding to the descending part of the body path.
- On the coordinate axes, indicate the special points corresponding to the selected solutions, i.e. v_0 and $v_\infty(v_0)$ on the ordinate axis and $x_*(v_0)$ on the abscissa axis, where appropriate.